Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	26 February 2026
Team ID	699d5f89c40b5ad0e9ee5981
Project Name	QUOTES RECOMMENDATION CHATBOT USING NLP
Maximum Marks	4 Marks

Technical Architecture:

The Quotes Recommendation Chatbot uses a layered architecture to process user queries and recommend relevant quotes. The system processes natural language input using NLP techniques, analyzes the sentiment or keywords, and retrieves matching quotes from a database. The application consists of a user interface, application logic for NLP processing, database storage for quotes, and optional external APIs for advanced language processing.

Reference :

Natural Language Toolkit (NLTK) Documentation
https://www.nltk.org/

Python Official Documentation
https://docs.python.org/3/

Kaggle Quote Dataset (for training data)
https://www.kaggle.com/datasets

Quotes Recommendation Chatbot Using NLP

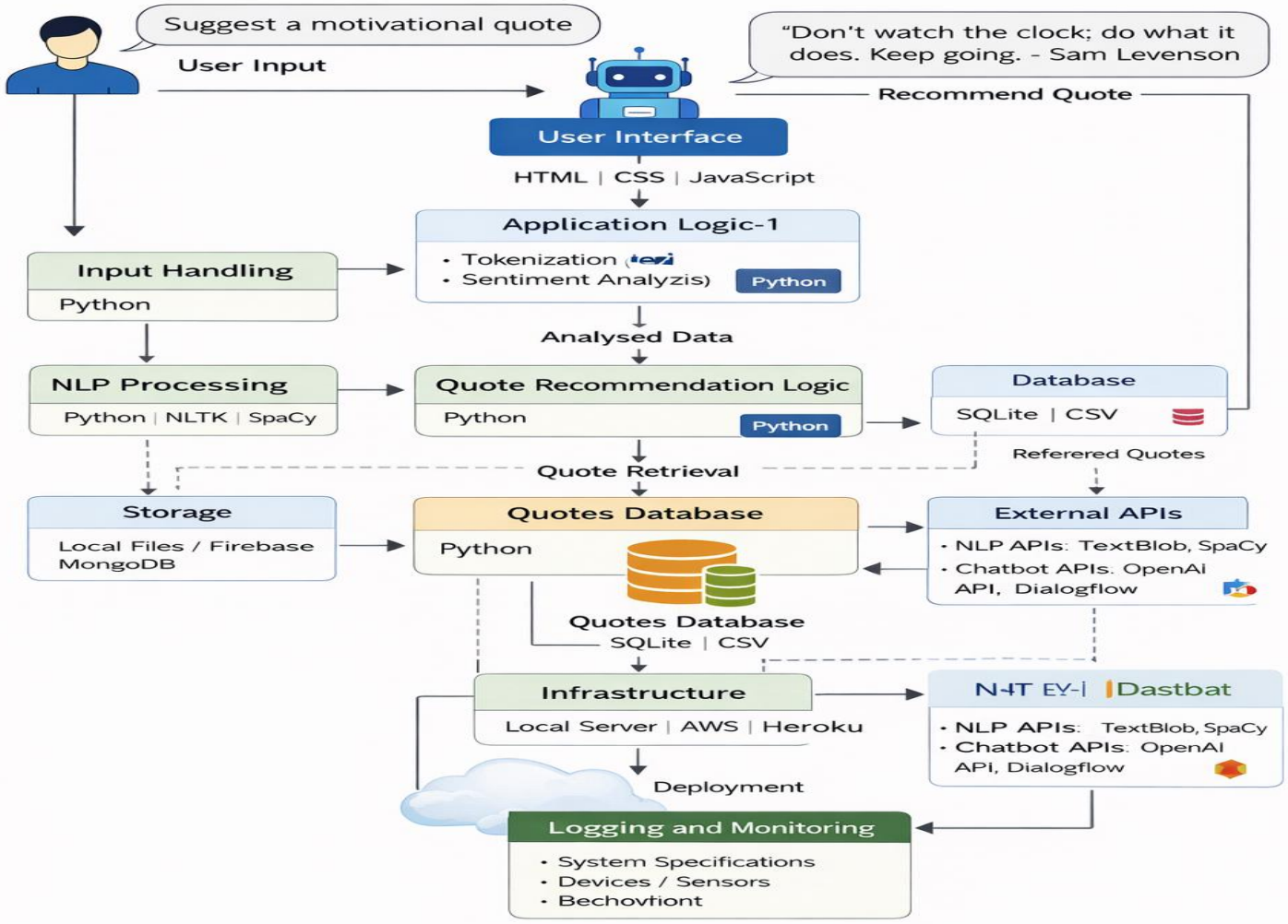

Table 1: Technology Stack Components

S.No	Component	Description	Technology
1.	User Interface	Interface where users interact with the chatbot and enter queries	HTML, CSS, JavaScript
2.	Application Logic-1	Handles user input and request processing	Python
3.	Application Logic-2	NLP processing such as tokenization, stop-word removal, and sentiment analysis	Python, NLTK
4.	Application Logic-3	Quote recommendation logic and response generation	Python
5.	Database	Stores quotes dataset including quote text, author, and category	MySQL, SQLite / CSV Dataset
6.	Cloud Database	Optional cloud storage for scaling the application	Firebase / MongoDB Atlas
7.	File Storage	Stores datasets, configuration files, and logs	Local Filesystem
8.	External API-1	NLP libraries used for text processing and sentiment analysis	NLTK / TextBlob
9.	External API-2	Optional APIs for advanced language processing or chatbot integration	OpenAI API / Dialogflow
10.	Machine Learning Model	Model used for sentiment analysis and text classification	NLP Sentiment Analysis Model
11.	Infrastructure (Server / Cloud)	Deployment environment where the application runs	Local Server / AWS / Heroku

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Frameworks used for developing the NLP chatbot system	Frameworks used for developing the NLP chatbot system
2.	Security Implementations	Security measures such as input validation and data protection	Security measures such as input validation and data protection
3.	Scalable Architecture	The system can be scaled by deploying on cloud infrastructure and expanding the quotes database	Cloud Deployment
4.	Availability	System designed to be available for users anytime through web or application interface	Web Server / Cloud Hosting
5.	Performance	System optimized to provide fast responses to user queries using efficient NLP processing and quick database retrieval.	Python Optimization, Indexed Dataset